Gold Eval — OCR & Extraction Pipelines

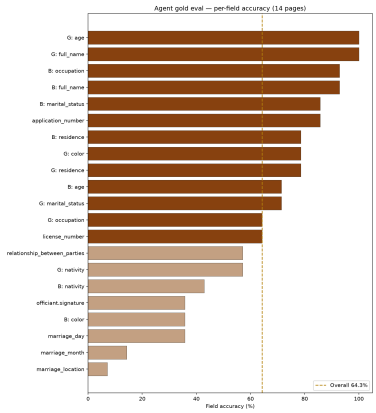
Baltimore marriage certificates · 14 labeled pages · 21 fields / page

Three pipelines on identical gold labels. **Docling + RapidOCR** and **Qwen2-VL 2B** feed OCR text into the same LangGraph extraction agent; **Qwen2.5-VL 7B** extracts structured fields **zero-shot** from the page image (no OCR text, no agent). Field accuracy rises 64.3% → 91.2% (**Qwen2.5-VL 7B leads by 26.9 pp**). Note the zero-shot model trades a few hallucinated header fields for far fewer missed values.

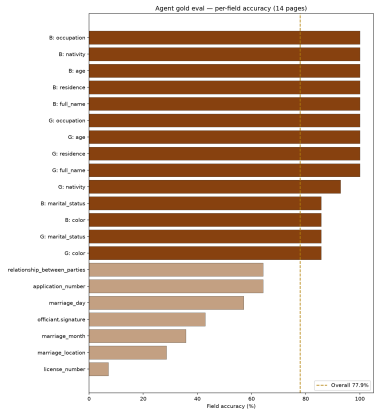
METRIC	DOCLING + RAPIDOCR OCR → agent	QWEN2-VL 2B OCR → agent	QWEN2.5-VL 7B zero-shot
Field accuracy	64.3%	77.9%	**91.2%**
Correct	189	229	**268**
Wrong	51	14	19
Missed	54	51	7
Hallucinated	0	0	14
Pages evaluated	14	14	14

PER-FIELD ACCURACY

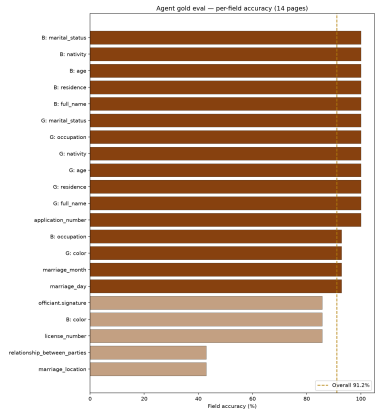
Docling + RapidOCR



Qwen2-VL 2B



Qwen2.5-VL 7B



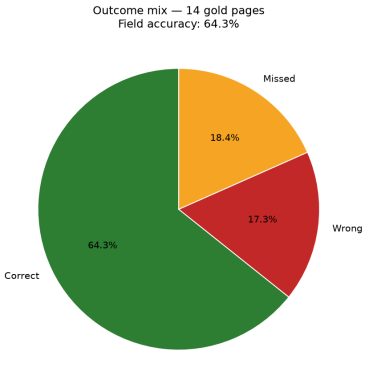
Runs — docling: docling-gold-results/plots/2026-07-05_135849 · vlm: vlm-gold-results/plots/2026-07-06_063003 · qwen25: qwen-2.5-vl-7b/plots/color-fix

Outcomes & per-page accuracy

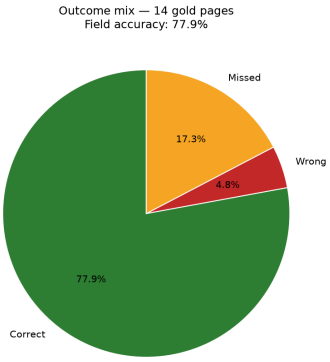
Correct / wrong / missed distribution, form-block accuracy, and page-by-page results

OUTCOME DISTRIBUTION

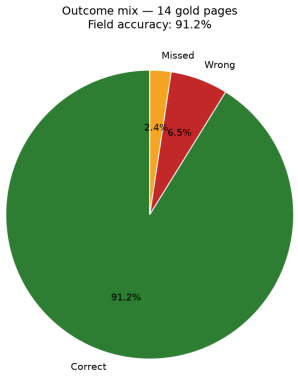
Docling + RapidOCR



Qwen2-VL 2B

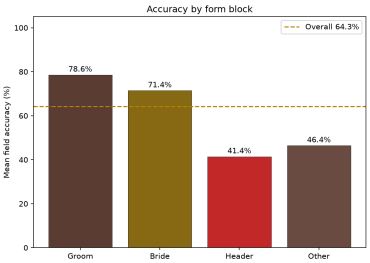


Qwen2.5-VL 7B

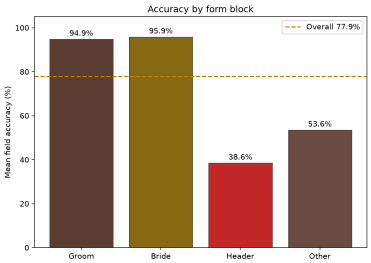


ACCURACY BY FORM BLOCK

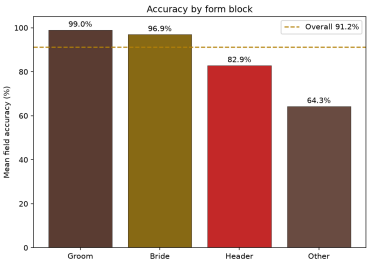
Docling + RapidOCR



Qwen2-VL 2B

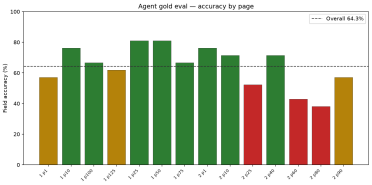


Qwen2.5-VL 7B

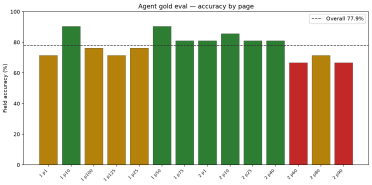


PER-PAGE FIELD ACCURACY

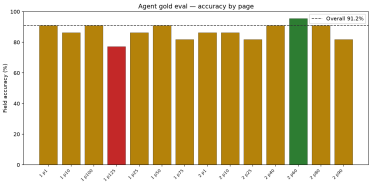
Docling + RapidOCR



Qwen2-VL 2B



Qwen2.5-VL 7B



Field-level comparison & errors

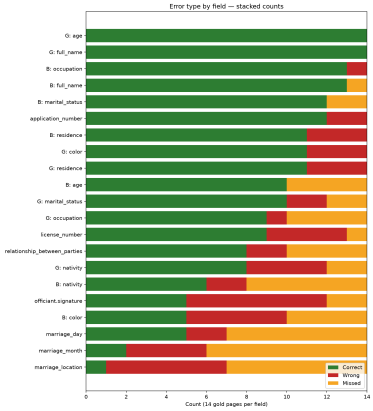
Per-field accuracy across all three pipelines and representative extraction errors

PER-FIELD ACCURACY — ALL PIPELINES

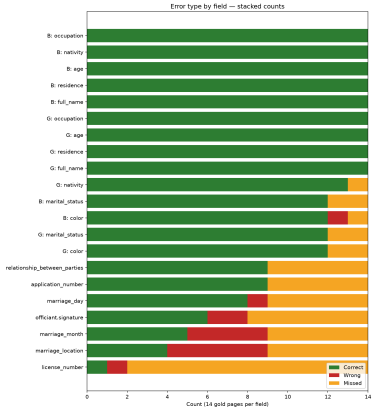
FIELD	DOCLING + RAPIDOCR	QWEN2-VL 2B	QWEN2.5-VL 7B
license_number	64.3%	7.1%	**85.7%**
application_number	85.7%	64.3%	**100.0%**
marriage_day	35.7%	57.1%	**92.9%**
marriage_month	14.3%	35.7%	**92.9%**
marriage_year	0.0%	0.0%	0.0%
marriage_location	7.1%	28.6%	**42.9%**
g: full_name	**100.0%**	**100.0%**	**100.0%**
g: residence	78.6%	**100.0%**	**100.0%**
g: age	**100.0%**	**100.0%**	**100.0%**
g: color	78.6%	85.7%	**92.9%**
g: nativity	57.1%	92.9%	**100.0%**
g: occupation	64.3%	**100.0%**	**100.0%**
g: marital_status	71.4%	85.7%	**100.0%**
b: full_name	92.9%	**100.0%**	**100.0%**
b: residence	78.6%	**100.0%**	**100.0%**
b: age	71.4%	**100.0%**	**100.0%**
b: color	35.7%	**85.7%**	**85.7%**
b: nativity	42.9%	**100.0%**	**100.0%**
b: occupation	92.9%	**100.0%**	92.9%
b: marital_status	85.7%	85.7%	**100.0%**
relationship	57.1%	**64.3%**	42.9%
officiant	35.7%	42.9%	**85.7%**

ERROR TYPE BY FIELD

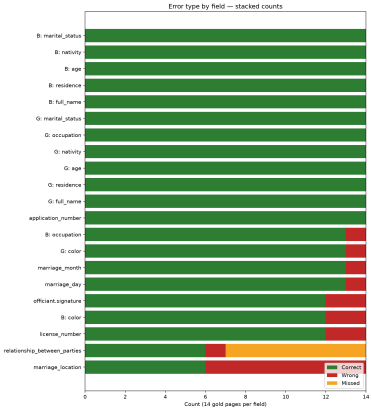
Docling + RapidOCR



Qwen2-VL 2B



Qwen2.5-VL 7B



REPRESENTATIVE ERRORS

Docling + RapidOCR · OCR → agent

PG	FIELD	TYPE	GOLD	PREDICTED
1 p1	marriage_location	MISSED	Balto	—

Qwen2-VL 2B · OCR → agent

PG	FIELD	TYPE	GOLD	PREDICTED
1 p1	license_number	MISSED	34178	—

Qwen2.5-VL 7B · zero-shot

PG	FIELD	TYPE	G
1 p1	marriage_year	HALLUCINATED	–

PG	FIELD	TYPE	GOLD	PREDICTED
1 p1	g: nativity	WRONG	TENN	TENor country)
1 p1	g: marital_status	MISSED	DIVORCED	—
1 p1	b: residence	WRONG	BALTO MD	DALTO D
1 p1	b: age	MISSED	23	—
1 p1	b: color	MISSED	COL	—
1 p1	b: nativity	MISSED	NY	—
1 p1	b: marital_status	MISSED	SINGLE	—
1 p1	relationship	MISSED	NONE	—
1 p10	marriage_month	WRONG	Aug	aug.

PG	FIELD	TYPE	GOLD	PREDICTED
1 p1	application_number	MISSED	5221	—
1 p1	marriage_day	MISSED	20	—
1 p1	marriage_month	MISSED	Aug	—
1 p1	marriage_location	MISSED	Balto	—
1 p1	relationship	MISSED	NONE	—
1 p10	license_number	MISSED	33892	—
1 p10	officiant	WRONG	Bert Constence	Bert Lonsley
1 p25	license_number	MISSED	33760	—
1 p25	application_number	MISSED	4766	—

PG	FIELD	TYPE	G
1 p1	relationship	MISSED	N
1 p10	marriage_year	HALLUCINATED	–
1 p10	marriage_location	WRONG	B
1 p10	relationship	MISSED	N
1 p25	marriage_year	HALLUCINATED	–
1 p25	marriage_location	WRONG	B
1 p25	b: occupation	WRONG	N
1 p50	marriage_year	HALLUCINATED	–
1 p50	relationship	MISSED	N

Generated from gold_eval_*.csv and plot PNGs in each run folder.